Exercise Sheet

Lecture 11

Dual Spaces, Functionals, and Multilinear Forms

The core is finite-dimensional and $V^* = \mathcal{L}(V, \mathbb{F})$ is the algebraic dual. A linear functional acts as $\varphi(v) = \varphi_i v^i$; the dual basis satisfies $e^i(e_j) = \delta^i_j$. With the physics convention, complex Riesz is conjugate-linear. A ⋆ marks a harder problem.

Core tutorial route

Work **11.1, 11.6, 11.11, 11.12, 11.13, 11.16, 11.24, and 11.25** in that order (about 85–100 minutes before discussion). This route covers a nonstandard dual basis, the finite annihilator dimension formula, pullback and transpose, composition reversal, the image–annihilator/rank proof, a bilinear calculation, finite complex Riesz versus the canonical double dual, and a non-diagonal covariance calculation with invariant evaluation.

Additional practice: 11.2–11.5, 11.7–11.10, 11.14–11.15, 11.17–11.18, and 11.20. **Bridge:** 11.21–11.23 (real metric and index lowering). **Extension:** 11.19 (alternating multilinear forms) and 11.26 (general algebraic double-dual injection). Bridge and extension tasks do not count toward core progress.

Dual basis and functionals

Exercise 11.1 (computational) Find the dual basis of $e_1 = (2, 1)$, $e_2 = (1, 1)$ in $\mathbb{F}^2$.

Answer. $e^1(x, y) = x - y$, $e^2(x, y) = -x + 2y$.

Exercise 11.2 (computational) On $\mathcal{P}_2$, express the functional $\varphi(a + bx + cx^2) = a + c$ in the dual basis of $1, x, x^2$.

Answer. $\varphi = e^0 + e^2$.

Exercise 11.3 (computational) Find the dual basis of $e_1 = (1, 0)$, $e_2 = (1, 1)$ in $\mathbb{F}^2$.

Answer. $e^1(x, y) = x - y$, $e^2(x, y) = y$.

Exercise 11.4 (computational) Find the dual basis of $e_1 = (3, 1)$, $e_2 = (5, 2)$ in $\mathbb{F}^2$.

Answer. $e^1(x, y) = 2x - 5y$, $e^2(x, y) = -x + 3y$.

Exercise 11.5 (computational) On $\mathcal{P}_2$ with basis $1, x, x^2$, expand $\varphi(a + bx + cx^2) = a - b + 2c$ in the dual basis and evaluate φ on $1 + 2x + 3x^2$.

Answer. $\varphi = e^0 - e^1 + 2e^2$; $\varphi(1 + 2x + 3x^2) = 5$.

Exercise 11.6 (proof) Let V be finite-dimensional. Prove that for a subspace $U \subseteq V$, $\dim U + \dim U^0 = \dim V$.

Exercise 11.7 (proof) Prove that the dual basis $e^1, \dots, e^n$ is linearly independent in V^*.

The dual map

Exercise 11.8 (computational) Let $T\colon \mathbb{F}^2 \to \mathbb{F}^2$ have matrix $\left(\begin{smallmatrix}2&1\\0&3\end{smallmatrix}\right)$. Compute $\mathcal{M}(T')$ in the dual bases.

Answer. $\mathcal{M}(T') = \mathcal{M}(T)^{\mathsf{T}} = \left(\begin{smallmatrix}2&0\\1&3\end{smallmatrix}\right)$.

Exercise 11.9 (computational) Let $T\colon \mathbb{F}^2 \to \mathbb{F}^2$ have matrix $\left(\begin{smallmatrix}1&2\\3&4\end{smallmatrix}\right)$. Compute $\mathcal{M}(T')$ in the dual bases.

Answer. $\mathcal{M}(T') = \left(\begin{smallmatrix}1&3\\2&4\end{smallmatrix}\right)$.

Exercise 11.10 (computational) Let $T\colon \mathbb{F}^3 \to \mathbb{F}^2$ have matrix $\left(\begin{smallmatrix}1&2&0\\0&1&3\end{smallmatrix}\right)$. Write $\mathcal{M}(T')$ and state the domain and codomain of T'.

Answer. $T'\colon (\mathbb{F}^2)^* \to (\mathbb{F}^3)^*$ with $\mathcal{M}(T') = \left(\begin{smallmatrix}1&0\\2&1\\0&3\end{smallmatrix}\right)$.

Exercise 11.11 (computational) Let $T\colon \mathbb{F}^2 \to \mathbb{F}^2$ have matrix $\left(\begin{smallmatrix}2&0\\1&3\end{smallmatrix}\right)$. For $\varphi = 2e^1 - e^2$ compute $T'\varphi$ in the dual basis.

Answer. $T'\varphi = 3e^1 - 3e^2$.

Exercise 11.12 (proof) Prove that $(ST)' = T'S'$ for $T\colon U \to V$, $S\colon V \to W$.

Exercise 11.13 (proof) Let $T\colon V \to W$ with V, W finite-dimensional. Prove the full dual-rank chain: $\operatorname{range} T' \subseteq (\operatorname{null} T)^0$, $\operatorname{null} T' = (\operatorname{range} T)^0$, $\operatorname{rank} T' = \operatorname{rank} T$, and hence $\operatorname{range} T' = (\operatorname{null} T)^0$. Explain why this is the dual-space proof that row rank equals column rank.

Exercise 11.14 (proof) Prove that $(S+T)' = S' + T'$ for $S, T\colon V \to W$.

Bilinear and multilinear forms

Exercise 11.15 (computational) Write the matrix of $B(u,v) = 2u_1v_1 - u_1v_2 + 3u_2v_2$, and split it into symmetric and antisymmetric parts.

Answer. Matrix $\left(\begin{smallmatrix}2&-1\\0&3\end{smallmatrix}\right)$; symmetric part $\begin{pmatrix}2&-\frac{1}{2}\\-\frac{1}{2}&3\end{pmatrix}$, antisymmetric part $\begin{pmatrix}0&-\frac{1}{2}\\\frac{1}{2}&0\end{pmatrix}$.

Exercise 11.16 (computational) For B with matrix $\left(\begin{smallmatrix}1&2\\3&1\end{smallmatrix}\right)$, $u = (1,1)$ and $v = (2,-1)$, compute $B(u,v) = u^{\mathsf{T}}Bv$.

Answer. $B(u,v) = 5$.

Exercise 11.17 (computational) Write the matrix of $B(u,v) = u_1v_1 + 4u_1v_2 + 2u_2v_1 - u_2v_2$ and split it into symmetric and antisymmetric parts.

Answer. Matrix $\left(\begin{smallmatrix}1&4\\2&-1\end{smallmatrix}\right)$; symmetric $\left(\begin{smallmatrix}1&3\\3&-1\end{smallmatrix}\right)$, antisymmetric $\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$.

Exercise 11.18 (computational) Write the matrix of the symmetric bilinear form on $\mathbb{R}^2$ whose quadratic form is $Q(v) = v_1^2 + 4v_2^2 + 2v_1v_2$.

Answer. $\left(\begin{smallmatrix}1&1\\1&4\end{smallmatrix}\right)$.

Exercise 11.19 (proof) Let $f\colon (\mathbb{F}^n)^n \to \mathbb{F}$ be multilinear in its n column arguments and suppose f vanishes whenever two columns are equal. Prove that f is then *alternating*: swapping two columns changes the sign. (The determinant is such an f, so this shows column swaps flip its sign.)

Exercise 11.20 (proof) Prove that every bilinear form B on V splits uniquely as $B = S + A$ with S symmetric and A alternating.

Covariant and contravariant

Exercise 11.21 (computational) In a basis with metric $g_{ij} = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right)$, lower the index of the vector with contravariant components $v^i = (1, 2)$, and compute $\langle v, v \rangle$.

Answer. $v_i = (2, 6)$; $\langle v, v \rangle = 14$.

Exercise 11.22 (computational) With metric $g_{ij} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 4 \end{smallmatrix}\right)$, lower the index of $v^i = (2, 1)$ and compute $\langle v, v \rangle$.

Answer. $v_i = (2, 4)$; $\langle v, v \rangle = 8$.

Exercise 11.23 (computational) With metric $g_{ij} = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, lower the index of $v^i = (1, 2)$ and compute $\langle v, v \rangle$.

Answer. $v_i = (3, 5)$; $\langle v, v \rangle = 13$.

Exercise 11.24 (computational) On $V = \mathbb{C}^2$ with the standard physics-convention inner product, let $\varphi(z_1, z_2) = (1 + i)z_1 - 2iz_2$ and $v = (2, i)$. Find the unique Riesz vector u with $\varphi = \langle u, \cdot \rangle$, verify $R(iu) = \bar{i}\, R(u)$, and compute both $\varphi(v)$ and $J(v)(\varphi)$ for the canonical $J\colon V \to V^{**}$.

Answer. $u = (1 - i, 2i)$; $R(iu) = -iR(u)$; and $\varphi(v) = J(v)(\varphi) = 4 + 2i$.

Exercise 11.25 (computational) Let $E' = EP$ for $P = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$. A fixed vector and fixed row covector have old components $[v] = (3, -1)^{\mathsf{T}}$ and $[\varphi] = (2, 5)$. Compute their new components, verify invariant evaluation, and state separately the matrix whose rows represent the new dual-basis elements in the old coordinates.

Answer. $[v]' = (7, -4)^{\mathsf{T}}$, $[\varphi]' = (7, 12)$, $[\varphi]'[v]' = [\varphi][v] = 1$; the new dual-basis rows form $P^{-1} = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right)$.

Exercise 11.26 ($\star$) Show that the canonical map $V \to V^{**}$, $v \mapsto (\varphi \mapsto \varphi(v))$, is injective for an arbitrary algebraic vector space (assuming the basis theorem), and hence is an isomorphism when V is finite-dimensional.

Hint. Given $v \neq 0$, produce a single functional that does not vanish on v: extend v to a basis and use the first dual covector.